

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

MATHEMATICS PAPER 19 | ANSWER KEY (Version A)

Unit 11 Basic Statistics (24) + Unit 10 Practical Geometry (15) + Unit 6, sections 6.1 and 6.2 (11)

Total Marks: 50

Unit 11: 24 | Unit 10: 15 | Unit 6: 11

Version A

Marking rules used on this paper.

- A missing **unit** loses $\frac{1}{2}$ **mark** in that part. It never cancels correct working.
- In Question 9 the construction is marked on the **arcs left on the page**. Lengths within **2 mm** and angles within **2°** get full marks.
- In Question 8 an arithmetic slip carried forward correctly into the next part loses the mark only once.

SECTION A: ANSWER KEY (10 × 1 = 10 MARKS)

1. Seconds in one minute of an angle: (c) 60 1

The book states $1^\circ = 60'$, $1' = 60''$ and $1^\circ = 3600''$. So one minute contains **60 seconds**. The distractor 3600 is the number of seconds in a **degree**, not in a minute.

Source: Unit 6, section 6.1.1 Sexagesimal System, book p115.

2. 1 radian is approximately: (a) $57^\circ 17' 45''$ 1

π radians = 180° , so 1 radian = $(180/\pi)^\circ \approx 57^\circ 17' 45''$. This exact value is printed in the book.

Source: Unit 6, section 6.1.2 Circular System or Radian Measure, book p117.

3. Area of a sector: (d) $A = \frac{1}{2}r^2\theta$ 1

The book derives $A = r^2\theta/2$, which is the same as $\frac{1}{2}r^2\theta$, where θ is in **radians**. Option (a) $r\theta$ is the **arc length**, (b) is the circumference and (c) is the area of the whole circle.

Source: Unit 6, section 6.2.2 Area of Sector of a Circle, book p119.

4. Where the three angle bisectors meet: (b) in-centre 1

The book lists the four points of concurrency: angle bisectors give the **in-centre** (inscribed centre), altitudes give the orthocentre, perpendicular bisectors give the circumcentre, and medians give the centroid.

Source: Unit 10, Key Fact p236 and KEY POINTS p237.

5. Right bisectors of a right-angled triangle: (c) mid point of the hypotenuse 1

The book gives all three cases: right bisectors of an acute triangle meet inside, of a right-angled triangle at the **mid point of the hypotenuse**, and of an obtuse triangle outside. Option (a), the vertex of the right angle, is where the **altitudes** of a right-angled triangle meet, which is the twin students most often pick.

Source: Unit 10, Note (a) to (c) on p234 and KEY POINTS p237.

6. Altitudes of an obtuse-angled triangle: (d) outside the triangle 1

The Key Fact box shows three figures: altitudes of an acute triangle meet inside, of a right-angled triangle at the vertex of the right angle, and of an obtuse triangle **outside** the triangle.

Source: Unit 10, Key Fact p233 and KEY POINTS p237.

7. Class mark of the class 5 to 15: (b) 10 1

Class mark is the average of the lower and upper limits: $(5 + 15) \div 2 = 10$.

Source: Unit 11, section 11.1 item (v) Class Marks / Mid Values, book p242.

8. The most frequent value: (c) mode 1

The book defines mode as "the most frequent (repeated) value of a data". Mean is the ratio of the sum to the number of values, and median is the middle most value of arranged data.

Source: Unit 11, section 11.2.1(c) Mode, book p254, and KEY POINTS p263.

9. Probability of an impossible event: (a) 0 1

Probability always lies between 0 and 1 inclusive. The book's scale marks an **impossible event at probability 0** and a certain or sure event at probability 1. Option (d) 2 is outside the range altogether.

Source: Unit 11, section 11.3 Probability scale, book p259, and Miscellaneous Exercise 11 Q1.

10. Expected number of heads in 400 tosses: (b) 200 1

Expected frequency = number of trials $\times$ probability of the event = $n \times p = 400 \times \frac{1}{2} = 200$.

Source: Unit 11, section 11.3.4 Expected Frequency, book p261.

SECTION B: ANSWER KEY ($6 \times 4 = 24$ MARKS)**2. Angle conversions 4****(i) 75° into radians, in terms of π . 2**

To convert degrees into radians, multiply by $\pi/180$.

$$75^\circ = 75 \times \pi/180 \text{ radians} = 75\pi/180 \text{ radians}$$

$$= 5\pi/12 \text{ radians}$$

1 mark for the rule or the substitution, 1 mark for $5\pi/12$ in lowest terms.

(ii) $5\pi/6$ radians into degrees. 2

To convert radians into degrees, multiply by $(180/\pi)^\circ$.

$$5\pi/6 \text{ radians} = (5\pi/6) \times (180/\pi)^\circ = 5 \times 180/6 = 150^\circ$$

1 mark for the rule or the substitution, 1 mark for 150° . The degree sign is part of the answer.

Source: Unit 6, section 6.1.2 conversion rules p117, worked Examples 5 and 6 p117 to p118, Exercise 6.1 Q5 and Q7.

3. Arc length and area of a sector, $r = 6$ cm, $\theta = \pi/3$ radians, $\pi = 3.14$ 4**(i) Length of the arc. 2**

$$l = r\theta$$

$$l = 6 \times \pi/3 = 2\pi = 2 \times 3.14$$

$$l = 6.28 \text{ cm}$$

(ii) Area of the sector. 2

$$A = \frac{1}{2}r^2\theta$$

$$A = \frac{1}{2} \times (6)^2 \times \pi/3 = \frac{1}{2} \times 36 \times \pi/3 = 6\pi = 6 \times 3.14$$

$$A = 18.84 \text{ cm}^2$$

In each part: 1 mark for the correct formula with substitution, 1 mark for the value. An answer left as 2π cm or 6π cm² is fully correct. Missing unit loses $\frac{1}{2}$ mark; cm instead of cm² in (ii) loses $\frac{1}{2}$ mark.

Source: Unit 6, section 6.2.1 p119 ($l = r\theta$), section 6.2.2 p119 ($A = \frac{1}{2}r^2\theta/2$), worked Examples 7 and 8 p120, Exercise 6.2 Q1.

4. Classes 5 to 14, 15 to 24, 25 to 34, 35 to 44. For the class 15 to 24 4**(i) Lower class limit = 15, upper class limit = 24. 1****(ii) Class boundaries. 1**

$$\text{Adjustment factor} = \frac{1}{2}(\text{lower limit of a class} - \text{upper limit of the preceding class}) = \frac{1}{2}(15 - 14) = 0.5$$

Subtract 0.5 from the lower limit and add 0.5 to the upper limit:

$$\text{Class boundaries} = 14.5 \text{ to } 24.5$$

(iii) Class mark. 1

$$\text{Class mark} = (15 + 24) \div 2 = 39 \div 2 = 19.5$$

(iv) Size of the class interval. 1

$$h = \text{difference of the lower limits of two consecutive classes} = 15 - 5 = 10$$

Accept h found from the upper limits, $24 - 14 = 10$, or from the boundaries, $24.5 - 14.5 = 10$.

Source: Unit 11, section 11.1 items (ii) Class Limits, (iii) Class Boundaries (p241), (iv) Size of Class Interval and (v) Class Marks (p242).

5. Mean, median and mode of 4, 6, 9, 6, 10, 7, 6, 12 4**(i) Mean.** 2

$$\Sigma x = 4 + 6 + 9 + 6 + 10 + 7 + 6 + 12 = 60, \quad n = 8$$

$$\bar{x} = \Sigma x / n = 60 / 8 = 7.5$$

(ii) Median. 1

Ascending order: 4, 6, 6, 6, 7, 9, 10, 12. Here $n = 8$, which is **even**.

$$\text{Median} = \frac{1}{2}\{(n/2)\text{th value} + ((n+2)/2)\text{th value}\} = \frac{1}{2}(4\text{th} + 5\text{th}) = \frac{1}{2}(6 + 7) = 6.5$$

(iii) Mode. 1

6 appears three times, more than any other value, so mode = 6.

In (ii), full marks only if the data is first arranged in order. Taking the middle of the unarranged list gives 6 and 10 and is wrong.

Source: Unit 11, Arithmetic Mean for Ungroup Data p250, Median for Ungroup Data p252 (Example 8 uses the same even- n rule), Mode p254.

6. Probability: 5 red, 3 green, 2 blue marbles 4

$$\text{Total number of outcomes} = 5 + 3 + 2 = 10$$

$$P(E) = \text{number of favourable outcomes} / \text{total number of outcomes}$$

$$\text{(i) } P(\text{green}) = 3/10$$
 1

$$\text{(ii) } P(\text{blue}) = 2/10 = 1/5$$
 1

$$\text{(iii) } P(\text{not red}) = 1 - P(\text{red}) = 1 - 5/10 = 5/10 = 1/2$$
 1

Also correct by direct counting: 3 green + 2 blue = 5 favourable out of 10.

$$\text{(iv) Expected number of 4s in 300 rolls}$$
 1

$$\text{Expected frequency} = n \times p = 300 \times 1/6 = 50 \text{ times}$$

Unsimplified fractions such as 2/10 and 5/10 are accepted in full.

Source: Unit 11, section 11.3.1 formula for $P(E)$ p260, section 11.3.2 complementary events p260, section 11.3.4 Expected Frequency p261.

7. Lines inside a triangle 4

(i) Altitude. 1 A **perpendicular line segment drawn from a vertex to the opposite side** of the triangle.

(ii) Median. 1 A **line segment joining the mid point of a side to the opposite vertex**.

(iii) The three altitudes meet at the orthocentre. 1

(iv) The centroid divides each median in the ratio 2 : 1. 1

In (iv) the book measures $PO : OB = 2 : 1$, the longer part being the one next to the vertex. Accept $2 : 1$. The book's own Miscellaneous Exercise 10 Q1(x) offers no $2 : 1$ option, so that printed item is faulty and is not used here.

Source: Unit 10, Altitudes of a Triangle p233, Medians of a Triangle p236, Key Fact p236 ($PO : OB = 2 : 1$), KEY POINTS p237.

SECTION C: ANSWER KEY ($2 \times 8 = 16$ MARKS)

8. Marks of 30 students: mean, median and mode 8**(i) The completed table.** 2

Marks	f	Class boundaries	Class mark x	fx	c.f.
1 to 10	4	0.5 to 10.5	5.5	22	4
11 to 20	6	10.5 to 20.5	15.5	93	10
21 to 30	10	20.5 to 30.5	25.5	255	20
31 to 40	6	30.5 to 40.5	35.5	213	26
41 to 50	4	40.5 to 50.5	45.5	182	30
Total	$\Sigma f = 30$			$\Sigma fx = 765$	

1 mark for correct class boundaries and class marks, 1 mark for the fx column with $\Sigma fx = 765$ and the c.f. column.

(ii) Arithmetic mean. 2

$$\bar{x} = \Sigma fx / \Sigma f = 765 / 30 = 25.5 \text{ marks}$$

(iii) Median. 2

$n/2 = 30/2 = 15$ th value. The c.f. column first passes 15 in the class 21 to 30, so the **median group is 20.5 to 30.5**.

Here $l = 20.5$, $h = 10$, $f = 10$, $n = 30$, $c = 10$ (the c.f. of the class **before** the median group).

$$\tilde{x} = l + (h/f)(n/2 - c) = 20.5 + (10/10)(15 - 10)$$

$$= 20.5 + 1 \times 5 = 25.5 \text{ marks}$$

The book's own "Danger zone" warning applies here: c is taken from the class before the median group, not from the median group itself.

(iv) Mode. 1

Highest frequency is 10, so the **modal class is 21 to 30**, boundaries 20.5 to 30.5.

$$l = 20.5, f_m = 10, f_1 = 6, f_2 = 6, h = 10$$

$$\text{Mode} = 20.5 + [(10 - 6) / (2 \times 10 - 6 - 6)] \times 10 = 20.5 + (4/8) \times 10$$

$$= 20.5 + 5 = 25.5 \text{ marks}$$

(v) The concluding sentence. 1

Any sentence with this sense earns the mark: "The mean, the median and the mode are all 25.5 marks, so the data is spread evenly (symmetrically) about the middle class 21 to 30."

Accept any wording that states all three are equal, or that the distribution is symmetrical. No mark if the sentence is missing.

Source: Unit 11, mean for grouped data p251, median for grouped data p253, mode for group data p254, empirical relation p256. Fresh data set, not taken from any book example.

9. Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 5$ cm, $CA = 4.5$ cm 8

(i) The triangle and the steps of construction. 3

- Draw $AB = 6$ cm with the ruler.
- With centre A and radius 4.5 cm, draw an arc above AB.
- With centre B and radius 5 cm, draw another arc cutting the first arc at C.
- Join C to A and C to B. ABC is the required triangle.

2 marks for the triangle drawn accurately with the two arcs still showing, 1 mark for the written steps. Lengths within 2 mm get full marks. If the arcs have been rubbed out, 1 mark is lost.

(ii) The three angle bisectors. 3

- With centre A and a convenient radius, draw an arc cutting AB at P and AC at Q.
- With centres P and Q and the same radius (more than half of PQ), draw two arcs meeting inside the triangle at R. Join A to R. AR is the bisector of $\angle A$.
- Repeat the same steps at B and at C to draw the bisectors of $\angle B$ and $\angle C$.
- The three bisectors cut each other at one point. Mark it O.

1 mark for each correctly bisected angle. Marks are for the compass arcs, not for a line drawn by eye with a protractor.

(iii) The conclusion. 2

Yes, the three angle bisectors are concurrent. They meet at a single point O, which is called the **in-centre** (inscribed centre) of the triangle, and it always lies **inside** the triangle.

1 mark for saying they are concurrent and naming the in-centre, 1 mark for saying it lies inside the triangle. A triangle with sides 6, 5 and 4.5 cm is acute angled, so O must come out inside; if her O lies outside, the construction is wrong.

Source: Unit 10, Bisectors of Angles of a Triangle and Example 6, book p232 (same construction, different side lengths), Key Fact p236, KEY POINTS p237. Exercise 10.2 Q1 asks exactly this.

Mark split for the report. Unit 11 Basic Statistics 24 | Unit 10 Practical Geometry 15 | Unit 6 sections 6.1 and 6.2 11. Unit 6 is the only part she has not been taught in class, so record it separately.